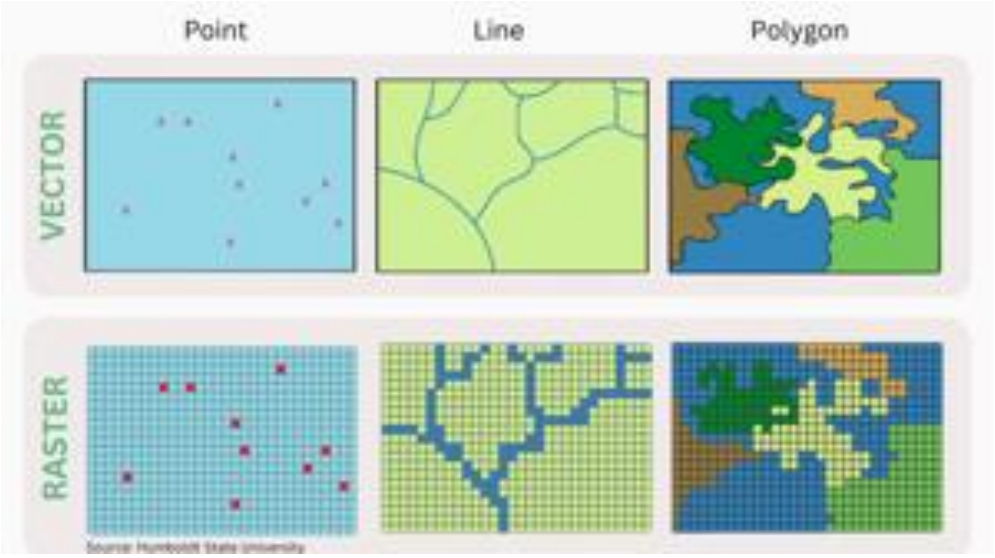
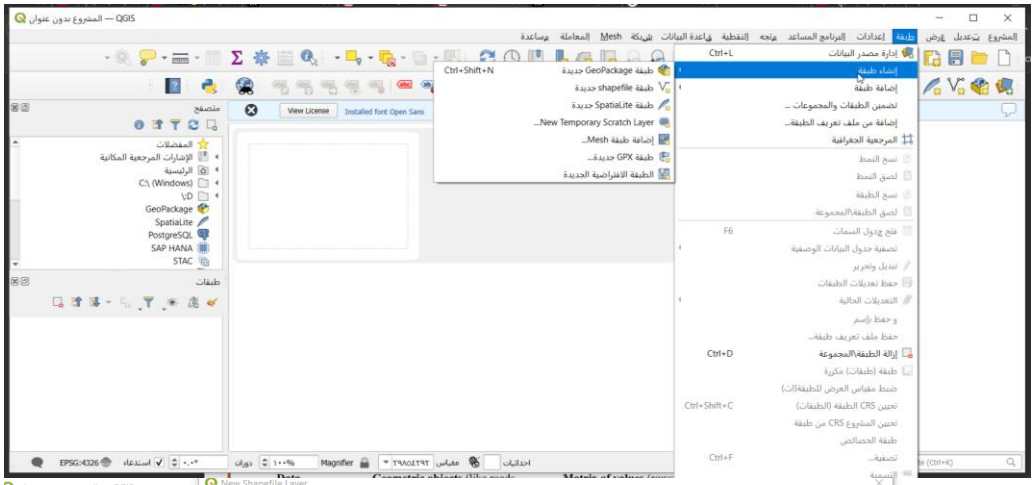
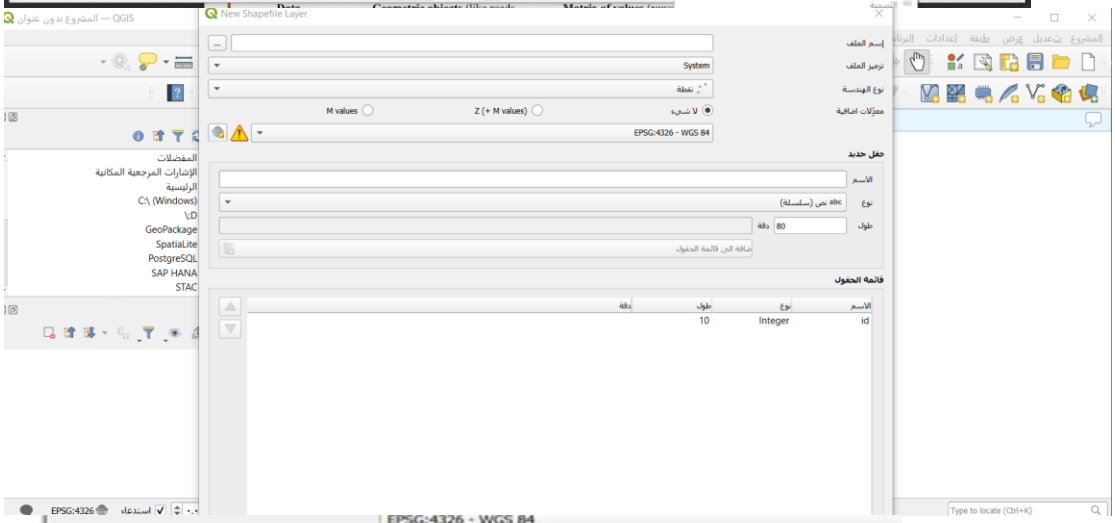
What is the key structural difference between Vector Data and Raster Data?

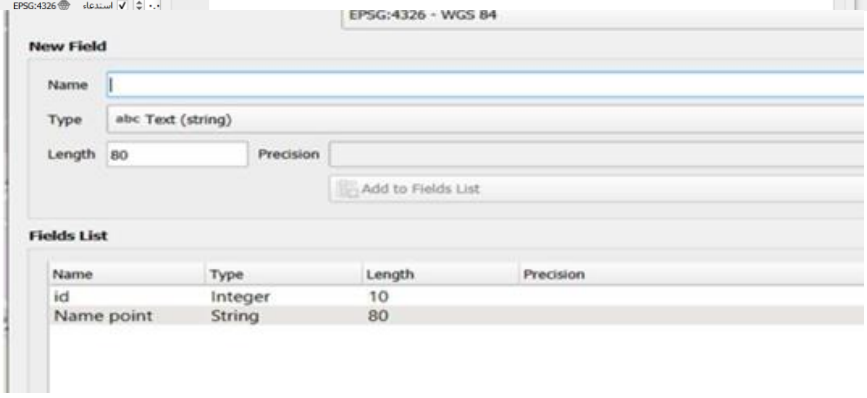


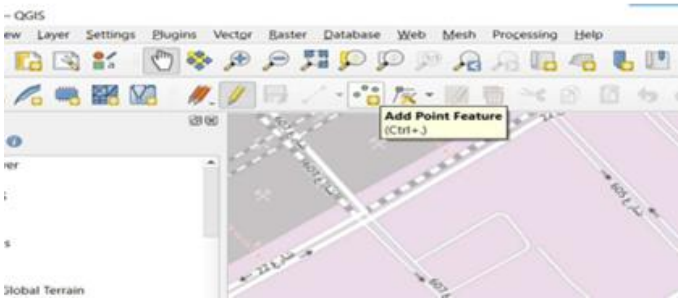
- Vector Data uses geometric shapes—points, lines, and polygons—defined by precise coordinate pairs to represent discrete spatial features like roads, boundaries, or buildings. Each feature can have rich attribute data associated with it.
- Raster Data represents space as a grid of uniform cells (pixels), where each cell holds a value representing a continuous variable such as elevation, temperature, or satellite imagery color bands. The data is inherently pixel-based and resolution-dependent.

Vector Point, line or polygon using QGIS:

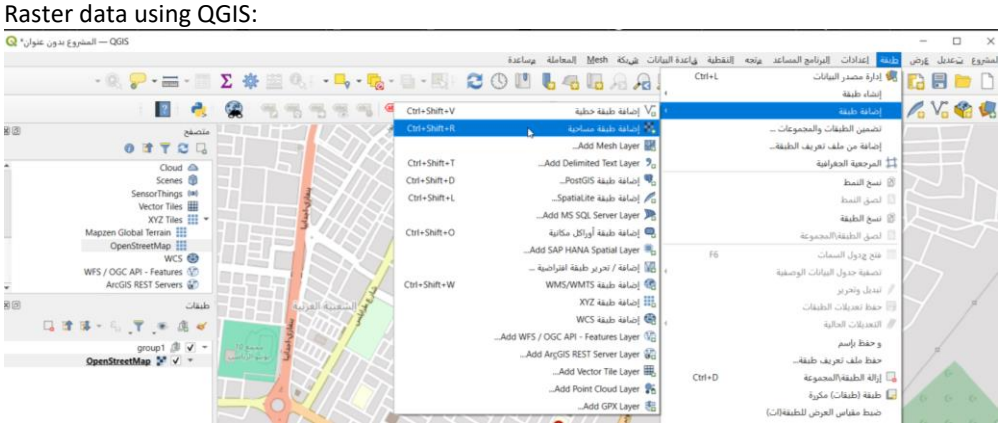

المشروع بدون عنوان — QGIS
طبقة
إدارة مصدر البيانات Ctrl+L
إنشاء طبقة
إضافة طبقة
تضمين الطبقات والمجموعات ...
إضافة من ملف تعريف الطبقة...
المرجعية الجغرافية
طبقة GeoPackage جديدة Ctrl+Shift+N
طبقة shapefile جديدة
طبقة SpatiaLite جديدة
New Temporary Scratch Layer...
إضافة طبقة Mesh...
طبقة GPX جديدة...
الطبقة الافتراضية الجديدة
View License
Installed font Open Sans
المفضلات
الإشارات المرجعية المكانية
الرئيسية
C:\ (Windows)
\:D
GeoPackage
SpatiaLite
PostgreSQL
SAP HANA
STAC
طبقات
EPSG:4326
Magnifier
إزالة الطبقة/المجموعة Ctrl+D
Ctrl+Shift+C
Ctrl+F


المشروع بدون عنوان — QGIS
New Shapefile Layer
اسم الملف
ترميز الملف
System
نوع الهندسة
نقطة
معدّلات اضافية
M values
Z (+ M values)
لا شيء
EPSG:4326 - WGS 84
حقل جديد
الاسم
نوع
نص (سلسلة)
طول
80
دقة
قائمة الحقول
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id
Integer
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المفضلات
الإشارات المرجعية المكانية
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C:\ (Windows)
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GeoPackage
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PostgreSQL
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STAC
EPSG:4326
Type to locate (Ctrl+K)


EPSG:4326 - WGS 84
New Field
Name
Type
abc Text (string)
Length
80
Precision
Add to Fields List
Fields List
Name
Type
Length
Precision
id
Integer
10
Name point
String
80


- QGIS
ew Layer Settings Plugins Vector Raster Database Web Mesh Processing Help
Add Point Feature
(Ctrl+.)
Global Terrain

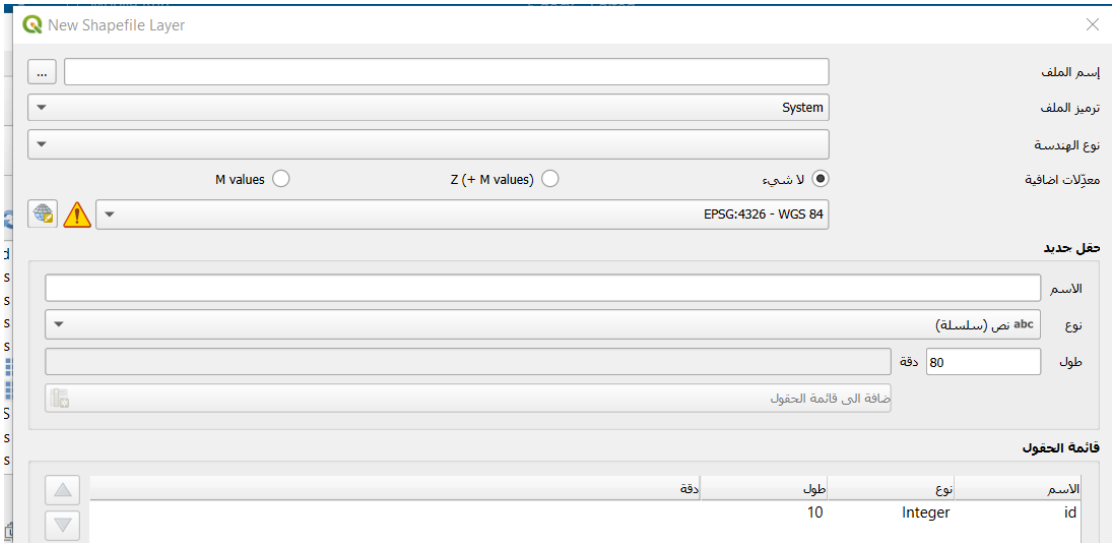
Raster data using QGIS:



Defining geometry and attributes -02 -03" hat is the meaning of this W-2:

When QGIS reports messages like **“Defining geometry”** and **“Attributes”**, it’s describing two fundamental steps in loading or processing a spatial layer:

1) **Defining geometry**
   This step tells you how QGIS is interpreting the spatial coordinates for each feature:
   For vector layers, it establishes the type of geometry (Point, LineString, Polygon) and the coordinate reference system (CRS).
   In a CSV or text import, it parses which fields correspond to X/Y (or latitude/longitude) and creates the actual spatial shapes.
   In short, it converts raw coordinate values into the spatial features you see on the map.
2) **Attributes**
   This step reads the non-spatial data (the “attribute table”) associated with each feature:
   These are columns such as names, IDs, measurements, categories, etc.
   QGIS loads all defined fields (string, integer, date, etc.) so you can display, query, style, or analyze based on those values.



Can you create any shapefile and change the symbology of this shapefile in different ways?

1. Create a New Shapefile Layer

In the QGIS main menu, choose **Layer ▶ Create Layer ▶ New Shapefile Layer...**.
In the dialog:

- Set **File name** to your desired path and name (e.g., my_points.shp).
- Choose **Geometry type**: Point (or LineString, Polygon).
- Click **Select CRS** to choose a coordinate reference system (e.g., WGS 84).
- In the **New Field** area, add attribute fields (for example:
  - id – Integer
  - name – Text (length 50)
  - value – Decimal number
  -). Click **Add to Fields List** after each.
- Click **OK**. QGIS adds an empty layer to the Layers panel.

To populate it, right-click the layer and choose **Toggle Editing**, then use the **Add Point Feature** tool. Click on the map to place points and fill in the attribute form. When finished, click **Save Edits** and **Toggle Editing** again.

Attributes

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2. Change the Symbology

Right-click your new layer and select **Properties ▶ Symbology**. Here are four common styling approaches:

A. Single Symbol

– The simplest style: all features share the same marker.

– Choose **Symbol layer type** (e.g., “Simple marker”), then adjust its shape (circle, square, triangle), size, fill color, stroke color, and width.

B. Categorized

– Style features based on a *categorical* attribute (e.g., name).

Change the top dropdown from “Single symbol” to **Categorized**.

Set **Value** to the field (e.g., name).

Click **Classify**: QGIS assigns a unique color to each distinct attribute value.

You can double-click any symbol in the list to tweak its color or marker shape.

C. Graduated

– Ideal for *numeric* attributes (e.g., value).

1. Switch to **Graduated**.
2. Select **Column**: choose the numeric field (value).
3. Choose a **Mode** (e.g., Equal Interval, Quantile, Natural Breaks).
4. Set **Classes** (number of ranges).
5. Pick a **Color ramp**.
6. Click **Classify**; QGIS generates a legend with unique symbology per range.

D. Rule-based

– For advanced styling with logical expressions.

1. Select **Rule-based** from the symbology dropdown.
2. Click **+** to add a rule: define a filter (e.g., "value" > 100) and assign a symbol.
3. Add multiple rules for different attribute ranges or combinations (e.g., "value" <= 100 AND "value" > 50).
4. Rules are evaluated in order; you can reorder or group them.

New Shapefile Layer

إسم الملف

ترميز الملف: System

نوع الهندسة: نقطة

معدّلات اضافية: لا شيء | Z (+ M values) | M values

EPSG:4326 - WGS 84

حقل جديد

الاسم

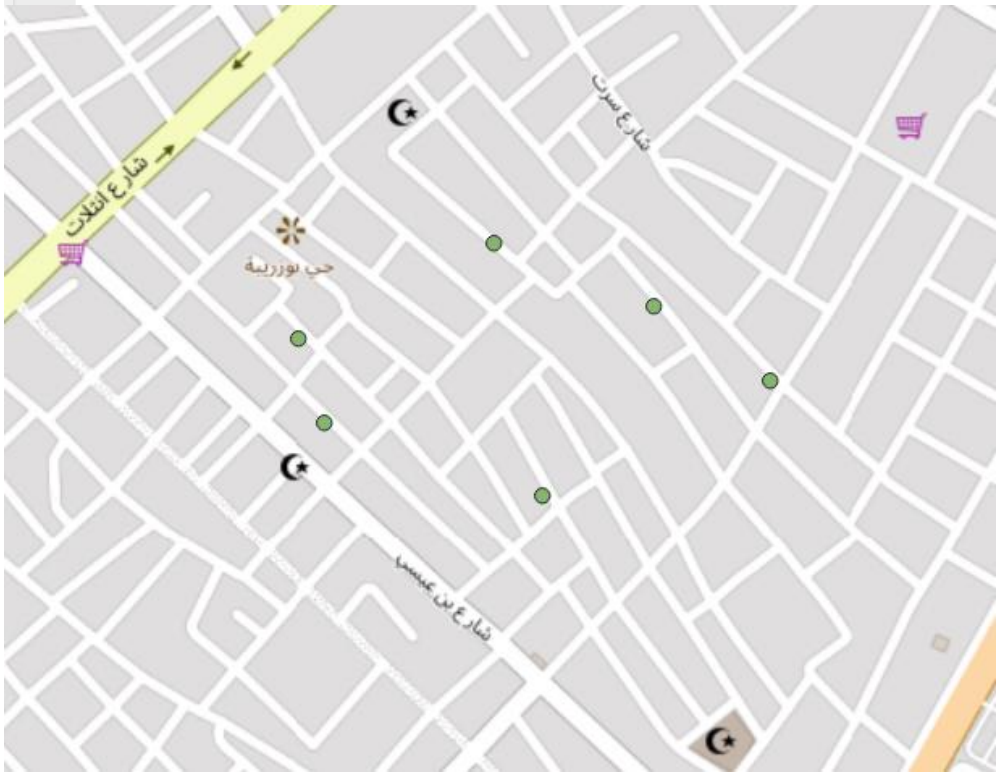
نوع: abc نص (سلسلة)

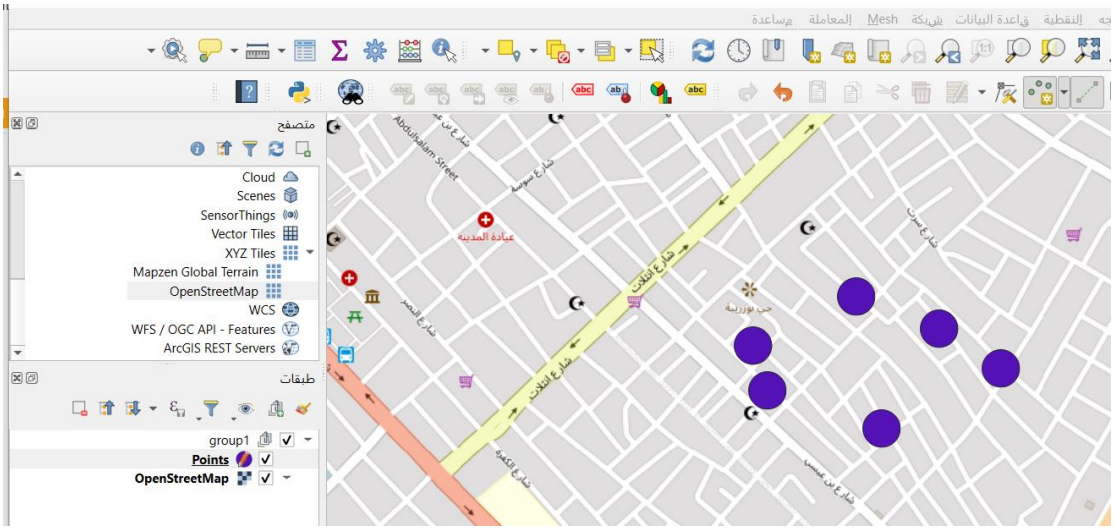
طول: 20 دقة: 6

اضافة الى قائمة الحقول

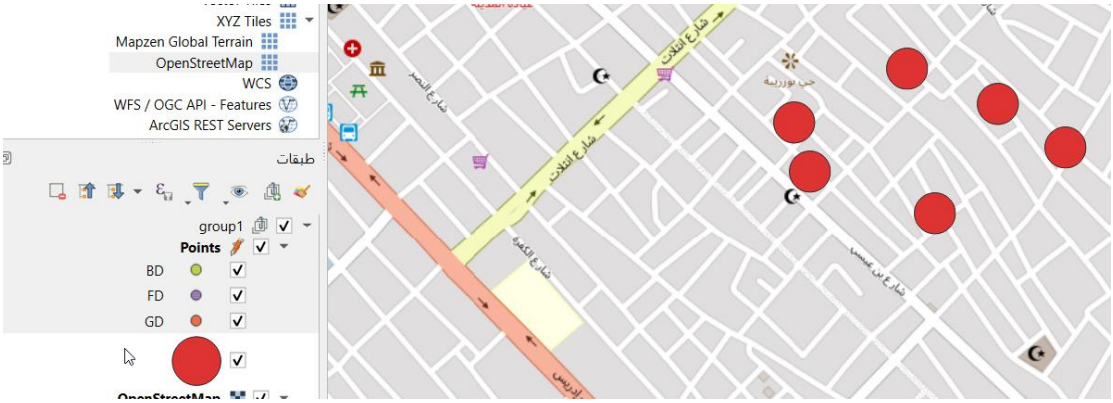
قائمة الحقول

الاسم	نوع	طول	دقة
id	Integer	10	
height	Real	20	6
width	Real	20	6
Name	String	20	




متصفح
Cloud
Scenes
SensorThings
Vector Tiles
XYZ Tiles
Mapzen Global Terrain
OpenStreetMap
WCS
WFS / OGC API - Features
ArcGIS REST Servers
طبقات
group1
Points
OpenStreetMap

OR


XYZ Tiles
Mapzen Global Terrain
OpenStreetMap
WCS
WFS / OGC API - Features
ArcGIS REST Servers
طبقات
group1
Points
BD
FD
GD

OR

GRADED

what is the primary purpose of an "Attribute Join" , and what is the one common field required between two tables to perform it successfully?

Primary Purpose of Attribute Join

The **primary purpose of an Attribute Join** in QGIS is to combine tabular data from two different sources by linking records that share matching values in a common field. This allows you to:

- **Enrich spatial data** with additional non-spatial information from external tables (like CSV files)
- **Merge attribute information** from two vector layers to create a comprehensive dataset
- **Append statistical or descriptive data** to geographic features for enhanced analysis and visualization
- **Create relationships** between datasets that don't initially contain all the information you need

For example, you might have a shapefile of administrative boundaries with basic information, and a separate CSV file containing population statistics. An attribute join lets you combine these to create maps showing population data by region.

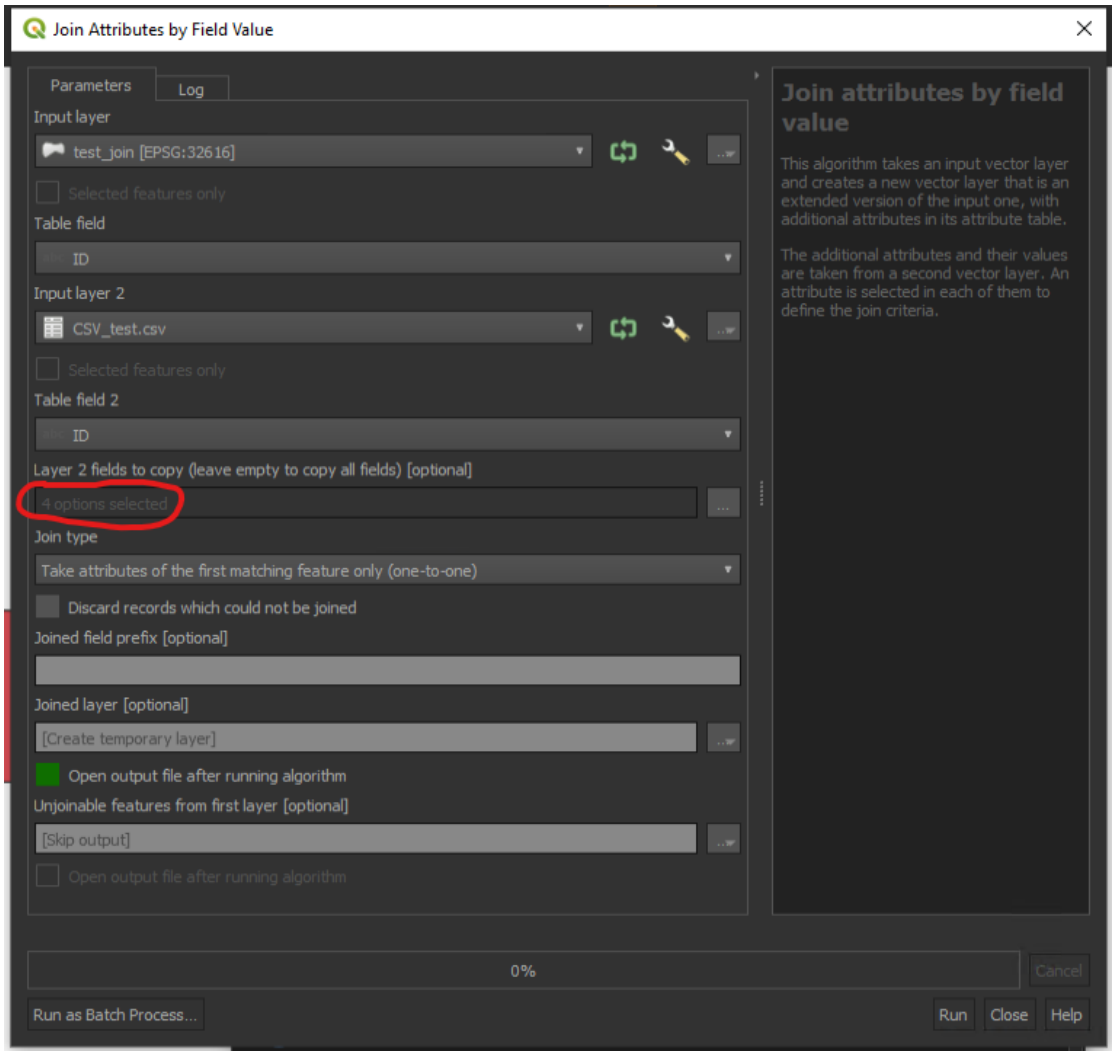
The One Common Field Required

To perform a successful attribute join, you need **one common field** (also called a **join key**) that exists in both tables with **matching values**. This field acts as the link between the two datasets and must meet these criteria:

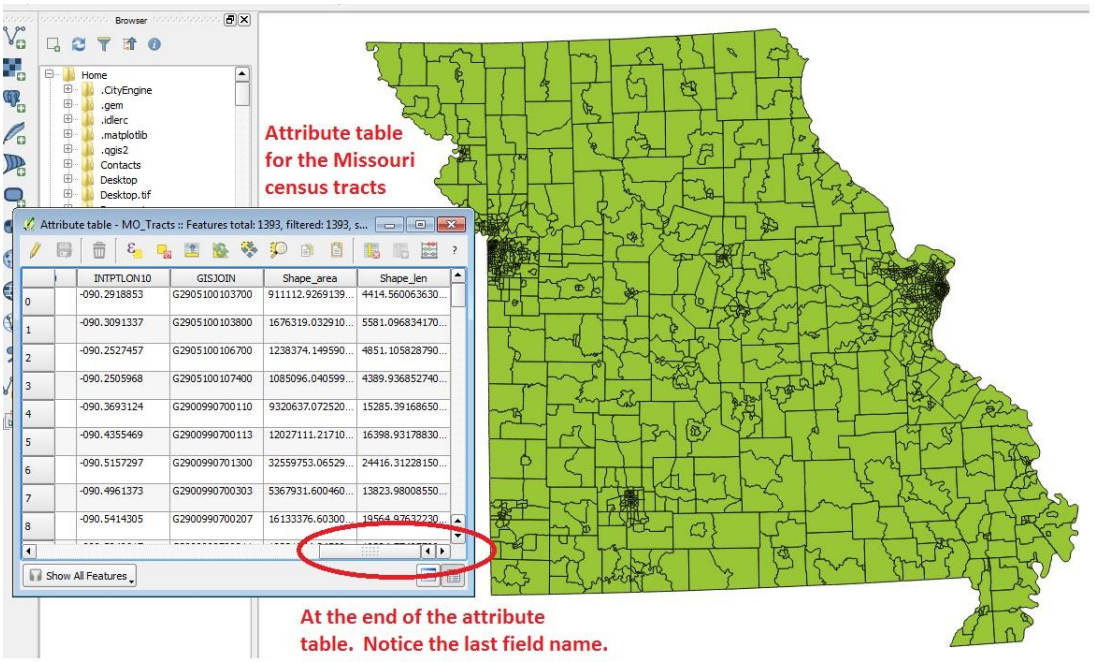
- **Same data type**: Both fields must be the same type (e.g., both integers, both text strings)
- **Matching values**: The field values in both tables must correspond to link the correct records
- **Ideally unique**: While not always required, having unique values prevents confusion and ensures accurate one-to-one matching

Common examples of join fields include:

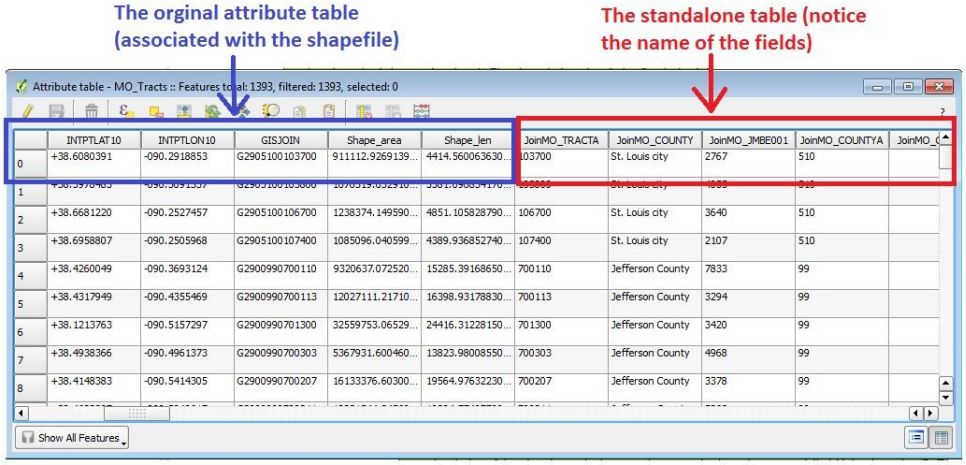
- **ID numbers** (e.g., GEOID, FID, ObjectID)
- **Names** (e.g., CountyName, StateName)
- **Codes** (e.g., PostalCode, AdminCode)



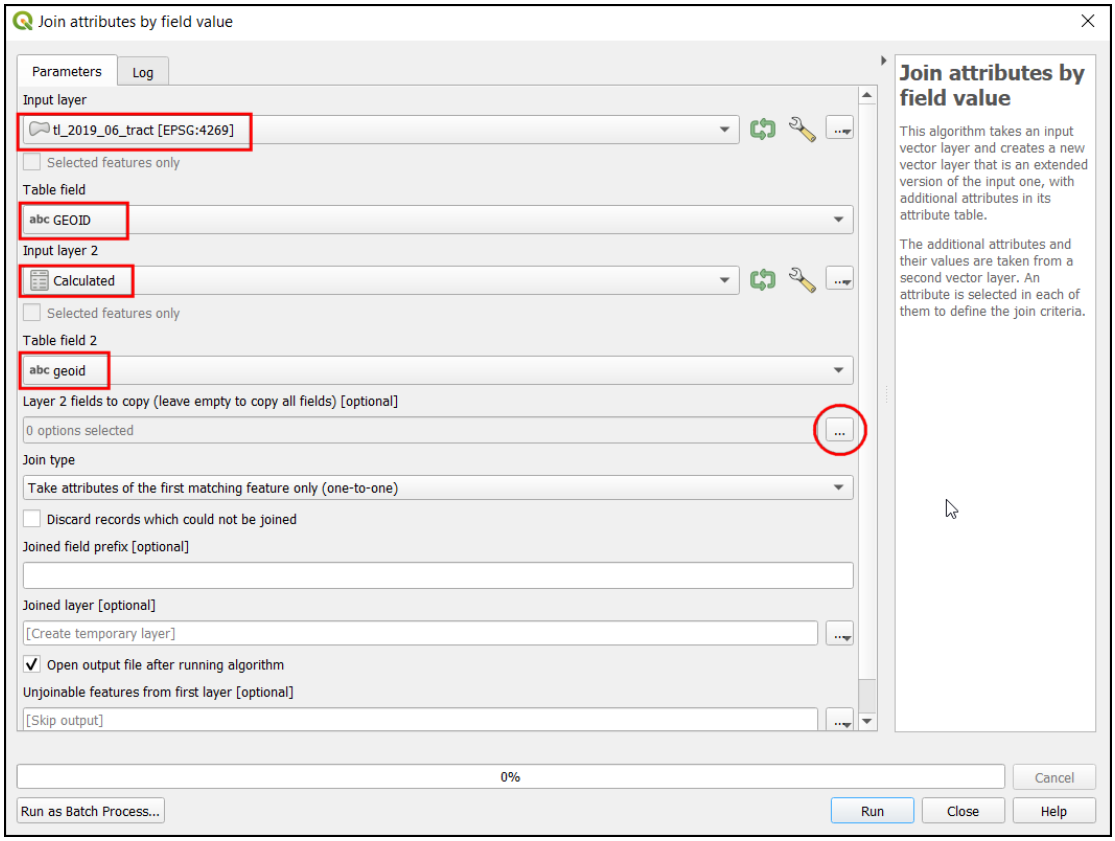
Shows the "Join Attributes by Field Value" tool dialog, demonstrating how to set up a join between two layers using common ID fields



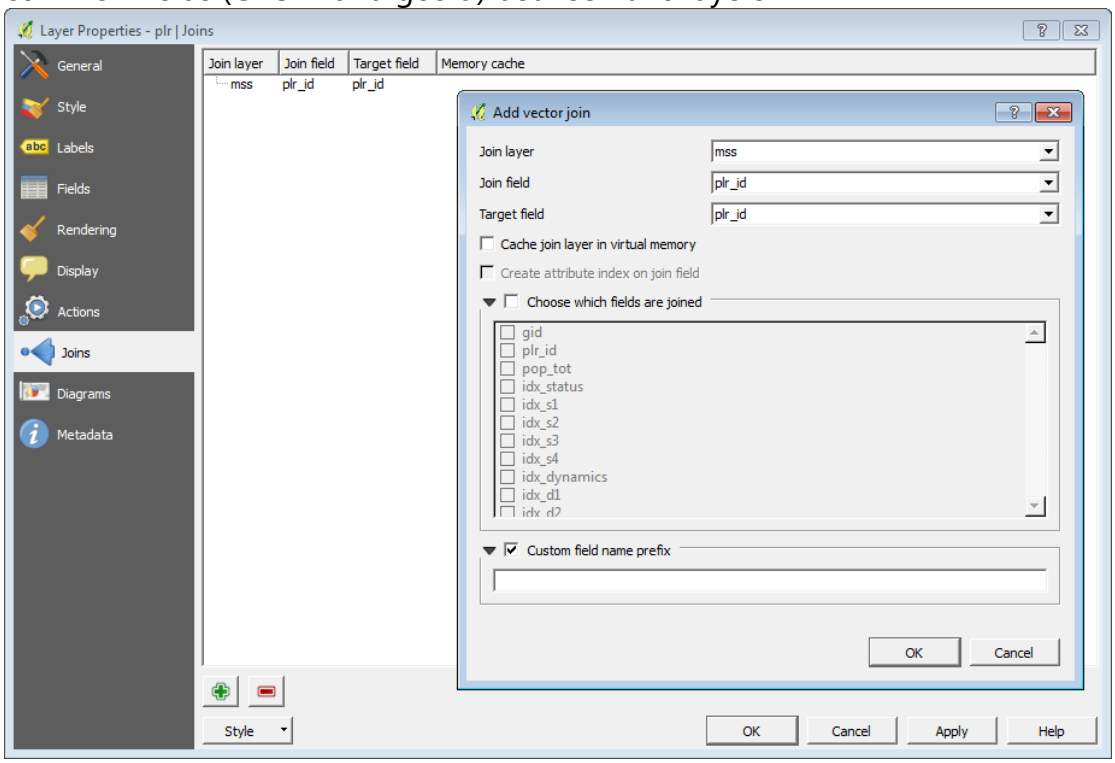
Once you join the table to the shapefile, you will notice the changes in the shapefile attribute table.



- Excellent before/after comparison showing a Missouri census tract attribute table before joining (highlighted in blue) and the new joined columns (highlighted in red) after the join operation



Another "Join attributes by field value" dialog showing the selection of common fields (GEOID and geoid) between two layers



The Layer Properties "Joins" tab interface, showing the "Add vector join" popup where you select the join layer and specify the common field (plr_id)